Prelim Report on BrPoliCorpus

Yearly Portuguese semantic-change quicklook on `BrPoliCorpus floor`

March 17, 2026

Bottom line. The **full-corpus yearly setup** is already producing interpretable political-semantic movement. The strongest preliminary evidence comes from *golpe*, *corrupção*, *previdência*, and *cpi*. The main remaining issue is **shortlist cleanliness**: this quicklook used the fast lookup-only preprocessing path, so the unrestricted top ranking still contains discourse markers and lemmatization artifacts.

1. Experimental Setup

Primary corpus	`BrPoliCorpus floor`
Window	2003–2023
Time slicing	Yearly for the main run; semester reserved for robustness
Scale	395,136 documents, 64.3M cleaned tokens, 3,558 scored lemmas

'BrPoliCorpus floor' here refers to **plenary floor speeches from the Câmara dos Deputados**. The speakers are primarily **federal deputies**, and each record includes the speech text together with exact date, session phase, speaker name, party, and state metadata. In practice, the corpus tracks how parliamentary language shifts across recurring legislative debates and public crises.

The embedding recipe follows the configuration of `nilc-nlp/word2vec-skip-gram-300d` as a **Portuguese embedding reference**. The pretrained NILC vectors are **not** used directly as the diachronic representation: the main experiment trains a separate Word2Vec Skip-Gram 300d model for each year and aligns the yearly spaces with Orthogonal Procrustes.

2. Why These Choices

`BrPoliCorpus floor` is the main corpus because it is the **largest and most continuous** Portuguese political-discourse panel currently organized for N2. Yearly slicing is the main granularity because it keeps each slice dense enough for stable embeddings while preserving politically meaningful temporal resolution. In this preliminary run, each yearly slice still contains **13.6k–26.9k documents per year** and **1.6M–4.7M cleaned words per year** (mean: 18.8k documents and about 3.1M cleaned words per year).

This makes yearly a better first-pass choice than coarser multi-year bins, which would blur major institutional shifts, and safer than semester slices for the first main experiment. Semester remains the first robustness check once the candidate list is cleaner.

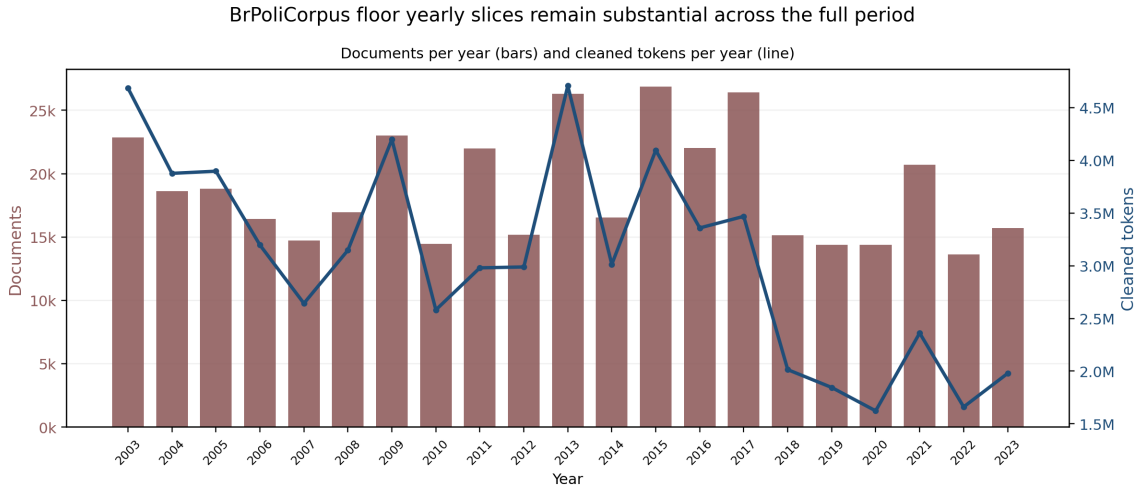


Figure 1: Yearly slice size remains substantial across the full 2003–2023 window.

3. Preliminary Results

The raw top ranking from this quicklook is not yet **paper-ready**, because the lookup-only preprocessing path still admits some noisy lemmas. For interpretation, the more useful view is the subset of politically meaningful terms that are frequent, present across the period, and supported by trajectory and neighborhood evidence.

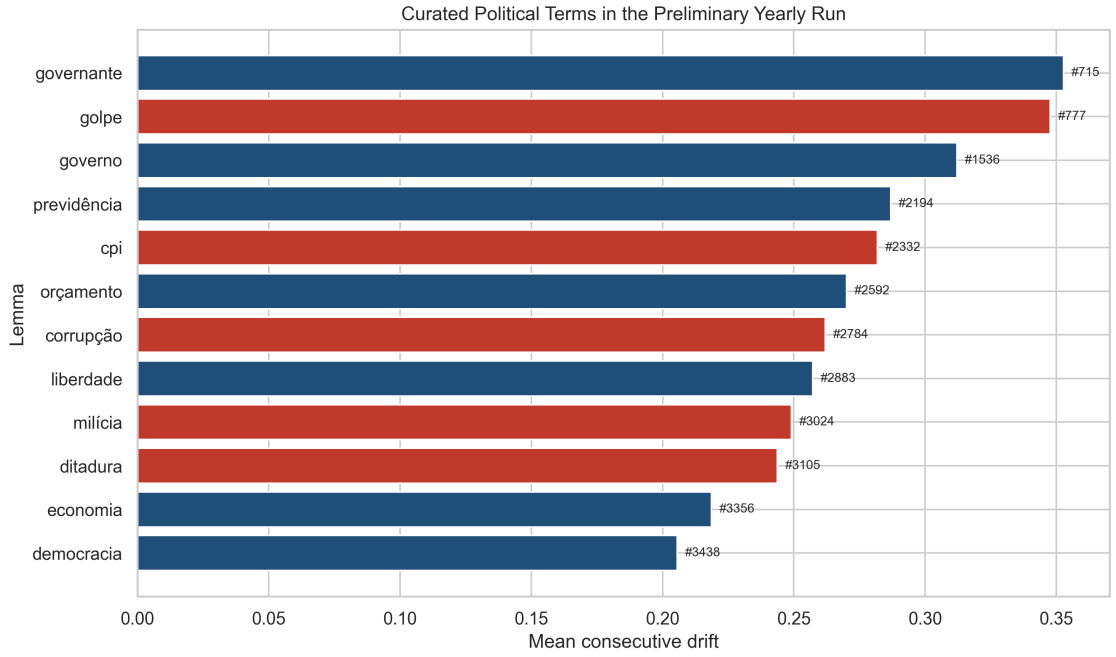


Figure 2: Politically interpretable terms in the preliminary yearly run.

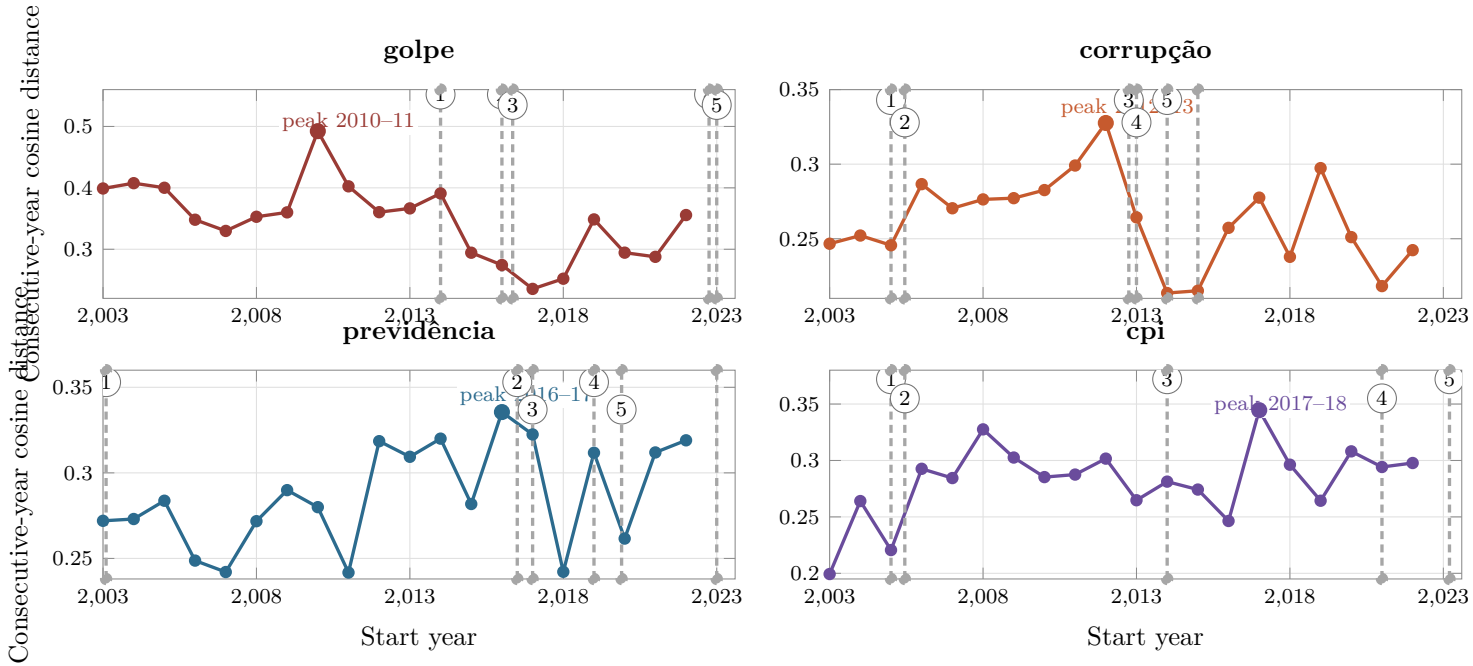
Table 1: Publication-style summary of four interpretable drift candidates. Neighbor sets are drawn from the 2003 and 2023 slice-specific models.

Term	Mean drift	Peak jump	2003 neighbors	2023 neighbors
golpe	0.348	2010→2011	desferir, exílio, golpista, dissolver, tacão	tramar, horda, golpista, golpismo, quebra-quebra
corrupção	0.262	2012→2013	malversação, roubalheira, corrupto, desmando, improbidade	esquema, peculato, lavagem, lava-jato, escândalo
previdência	0.287	2016→2017	previdenciário, resseguro, capitalização, atuarial, seguridade	previ, previdenciários, inativos, aposentadoria, assistencial
cpi	0.282	2017→2018	investigar, cpis, banestado, cpmi, apurar	cpmi, investigar, cpis, factoide, indiciamento

- **golpe** shows the **clearest shift**. Its 2003 neighborhood is broader and more historical (*desferir*, *exílio*, *golpista*), while 2023 is much more explicitly tied to anti-democratic rupture (*golpismo*, *quebra-quebra*, *tramar*). This is historically consistent with impeachment-era coup discourse and, especially, with the post-election and January 8 context.
- **corrupção** moves from a **general oversight/malfeasance frame** (*malversação*, *roubalheira*, *improbidade*) toward a more specific scandal and judicial-investigative frame (*lava-jato*, *petrolão*, *peculato*, *lavagem*). Its strongest movement is concentrated around 2011–2013, which fits the transition from the Mensalão period into the Lava Jato cycle.
- **previdência** remains a **policy-heavy term**, but the center of gravity changes. Early usage is anchored in generic reform and actuarial vocabulary, while late usage is more technical and institutional (*inativos*, *aposentadoria*, *previdenciários*). Its largest jump occurs in 2016–2017 and remains active into the 2019 reform cycle.
- **cpi** stays linked to investigation throughout, but its later neighborhood suggests a **more routinized accountability arena** rather than only scandal shorthand. The trajectory spikes around 2017–2018 and again later in the period, which is compatible with the renewed prominence of the CPI da Pandemia and the CPMI do 8 de Janeiro.

democracia moves much less than the four terms above. That is still useful: it may work better as a semi-stable comparison term than as a headline drift word.

The timing of the strongest movement is also **historically interpretable**. Figure 3 adds term-specific public-event anchors selected from official Câmara and Senado material used in the background check. These labels are meant to guide reading of the trajectories, not to claim direct causality.



golpe	**corrupção**
1 2014: Câmara marks 50 years of the 1964 coup.	**1** 2005: Correios scandal enters congressional agenda.
2 2016: Senate opens the Dilma impeachment process.	**2** 2005: Mensalão / Correios investigations consolidate.
3 2016: Dilma publicly frames impeachment as “golpe”.	**3** 2012: Mensalão trial intensifies corruption framing.
4 2022: Post-election encampments renew coup vocabulary.	**4** 2013: Senate approves corruption as a heinous crime.
5 2023: January 8 attack foregrounds anti-democratic rupture.	**5** 2014: Petrobras CPI and Lava Jato expand judicial framing.
previdência	**cpi**
1 2003: EC 41 reform redefines public-sector pensions.	**1** 2005: CPI dos Correios raises the salience of *cpi*.
2 2016: PEC 287 becomes a central reform dispute.	**2** 2005: Mensalão / Compra de Votos CPMI widens usage.
3 2017: Fiscal debate around PEC 287 peaks in Congress.	**3** 2014: Petrobras CPI restores the term’s national prominence.
4 2019: PEC 6 advances the Bolsonaro reform cycle.	**4** 2021: CPI da Pandemia brings the term back to center stage.
5 2019–2023: EC 103 and centenary debate renew the policy frame.	**5** 2023: CPMI do 8 de Janeiro refreshes the investigative frame.

Figure 3: Event-anchored yearly drift trajectories for four interpretable terms. Dashed lines show numbered public-event anchors; the key below the plots gives the corresponding episode for each term. Higher values indicate larger movement from year t to year $t+1$.

4. Current Status

This quicklook is already strong enough to support the project direction, but it should still be treated as **preliminary evidence rather than the final result**. The main reasons are simple: this pass used lookup-only preprocessing, only one replicate, and only five epochs. That is sufficient for a fast full-corpus diagnostic, but not yet for freezing the final shortlist.

The clean rerun with `pt_core_news_sm` should determine the candidate list used for the paper, the semester robustness check, and the confirmatory contextual stage. Even so, the current output already establishes **three important points**: the yearly design is viable, the corpus has enough density for diachronic modeling, and several politically meaningful terms already display interpretable semantic movement.